Lesson 05 Demo 02

Creating a React Application Using the useContext Hook

Objective: To develop a React application using Vite that demonstrates the functionality of the useContext Hook by implementing a theme toggle feature, allowing users to switch between light and dark themes dynamically

Tools required: Visual Studio Code and Node.js

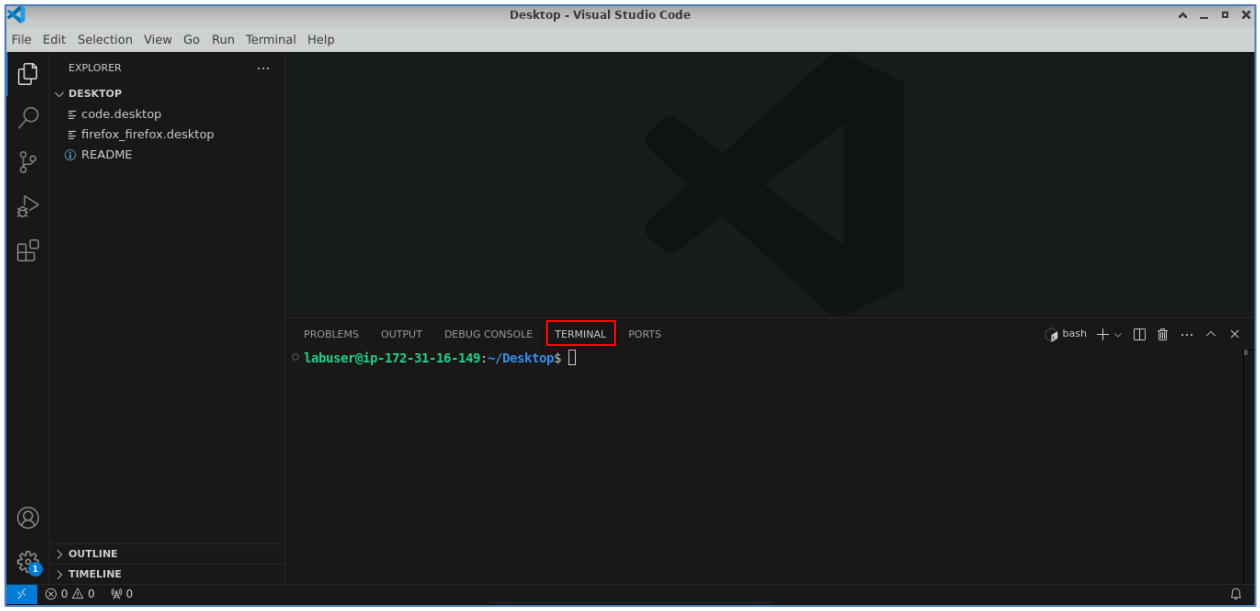
Prerequisites: None

Steps to be followed:

1. Set up a new React project using Vite
2. Create react components
3. Import and use the components
4. Run and verify the application

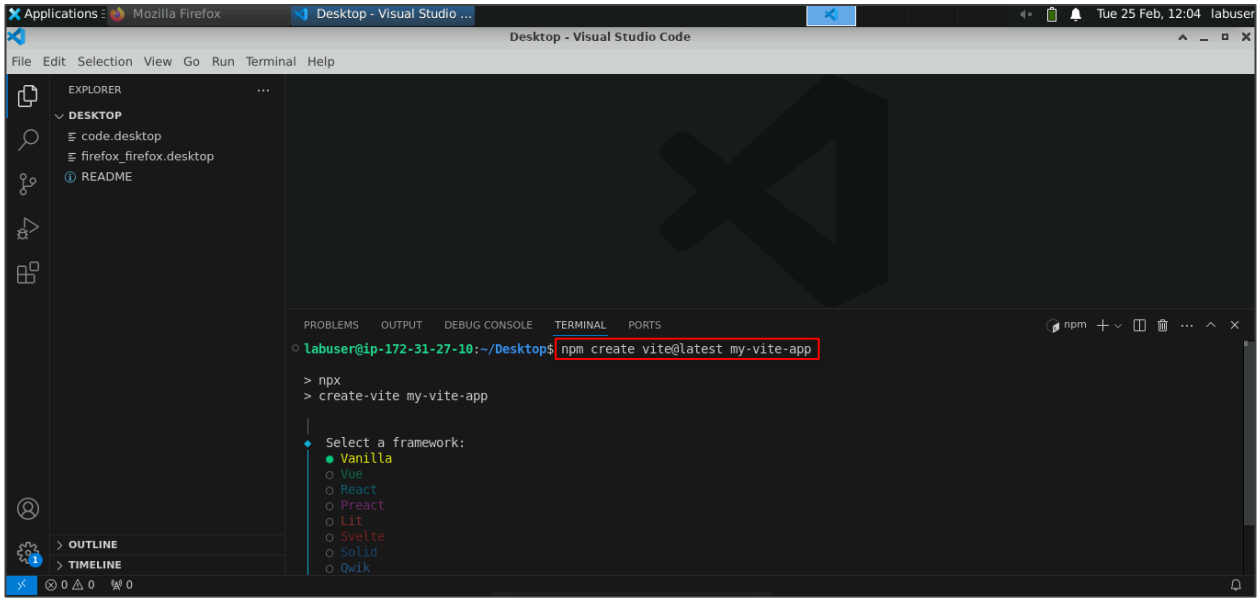
Step 1: Set up a new React project using Vite

1.1 Open Visual Studio Code (VS Code) and launch the terminal

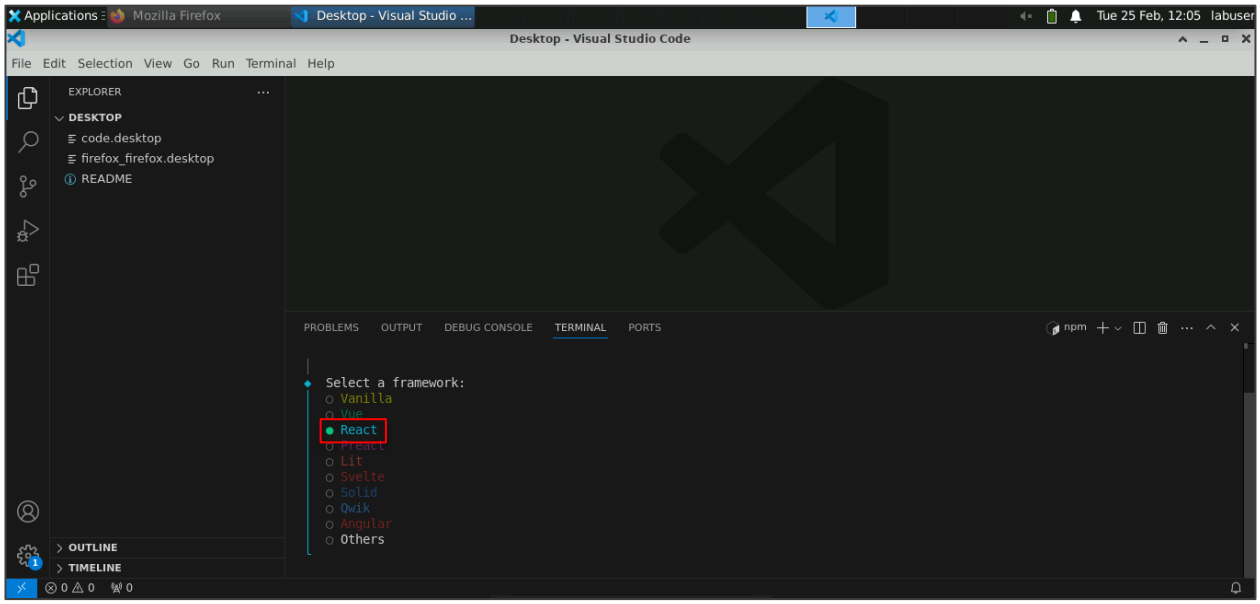


1.2 Run the following command to create a new React project using Vite:

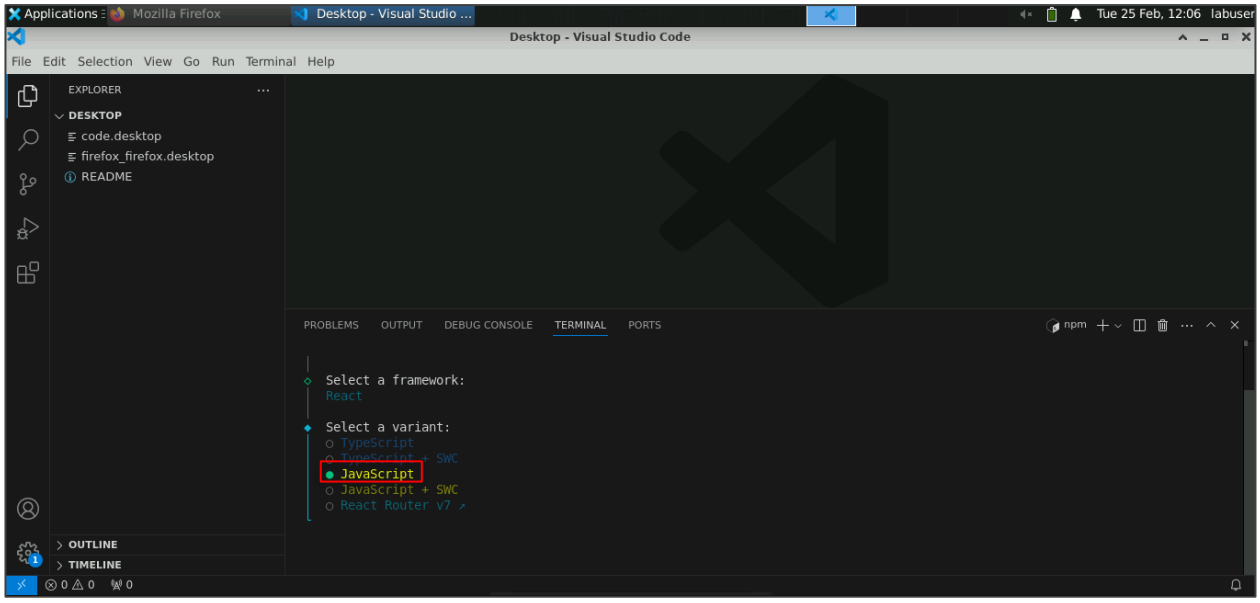
npm create vite@latest my-vite-app



1.3 Select **React** as the project framework and press enter

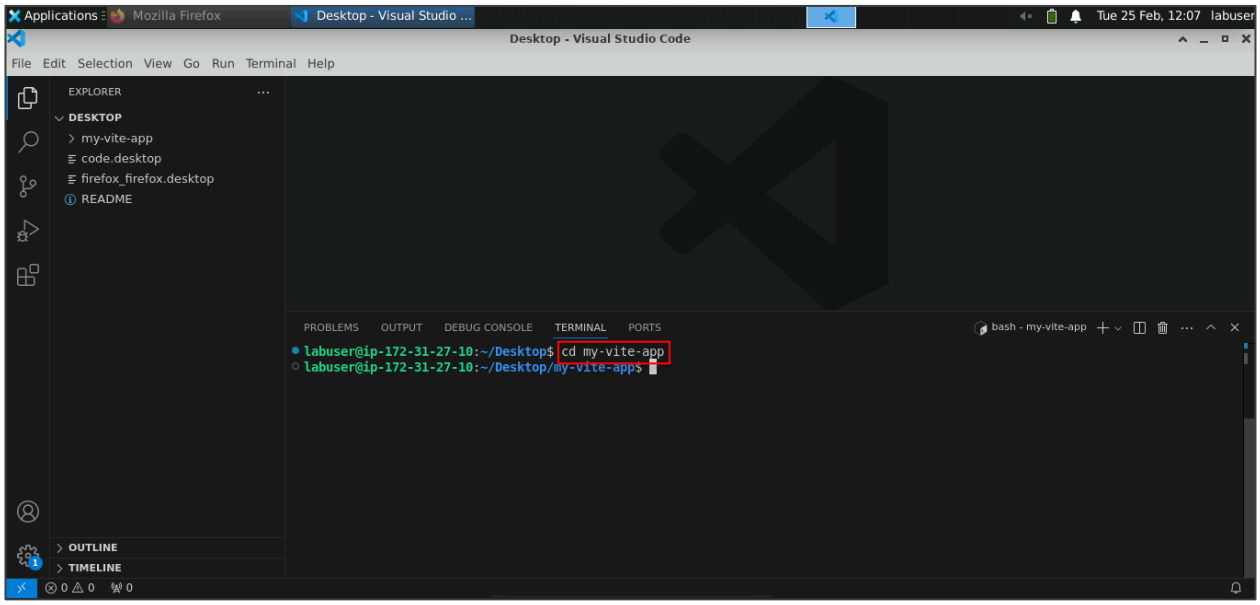


1.4 Select **JavaScript** as the language and press enter



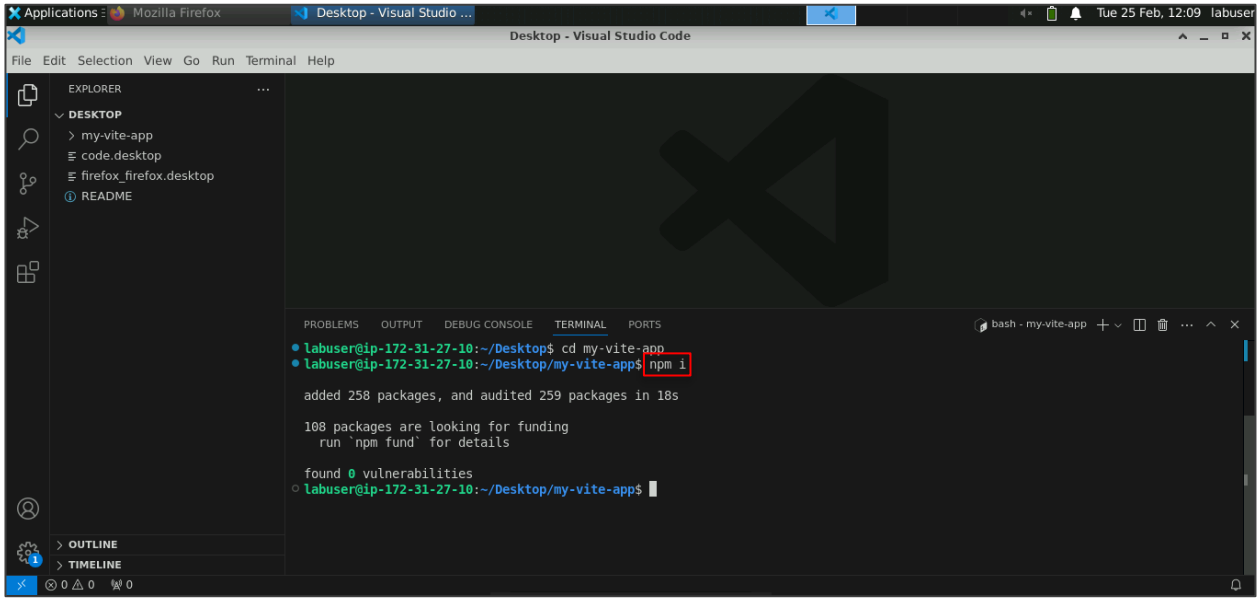
1.5 Run the following command to navigate to the project folder:

cd my-vite-app



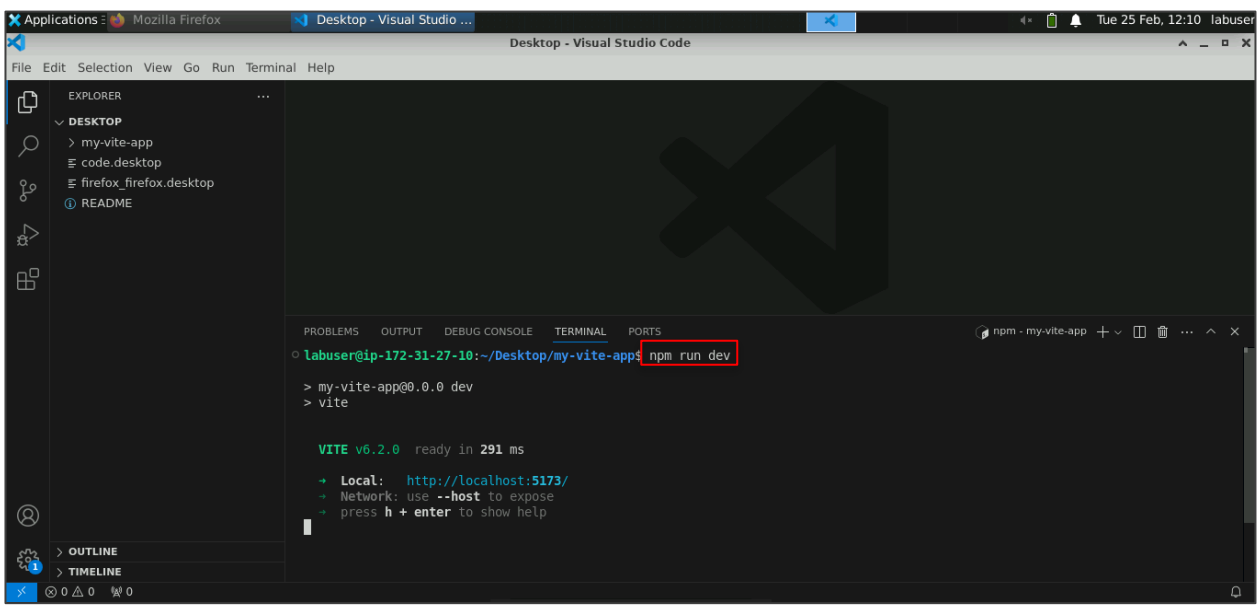
1.6 Run the following command to install dependencies:

npm i

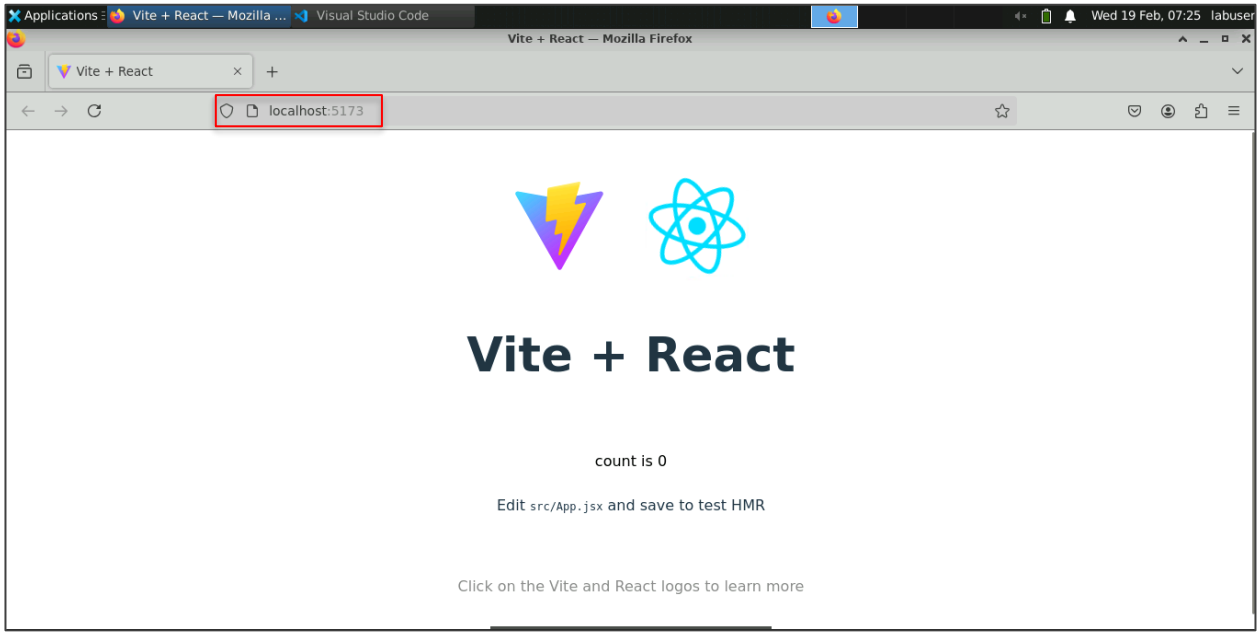


1.7 Run the following command to start the React development server:

npm run dev

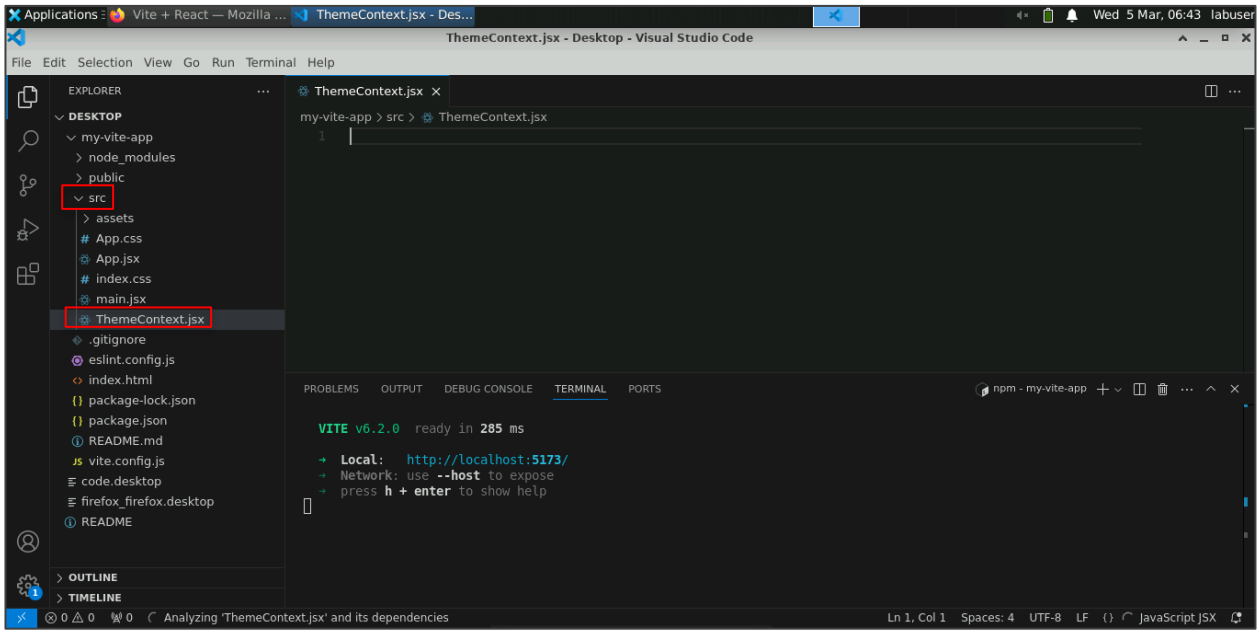


1.8 Navigate to the web browser and paste the following link:
http://localhost:5173



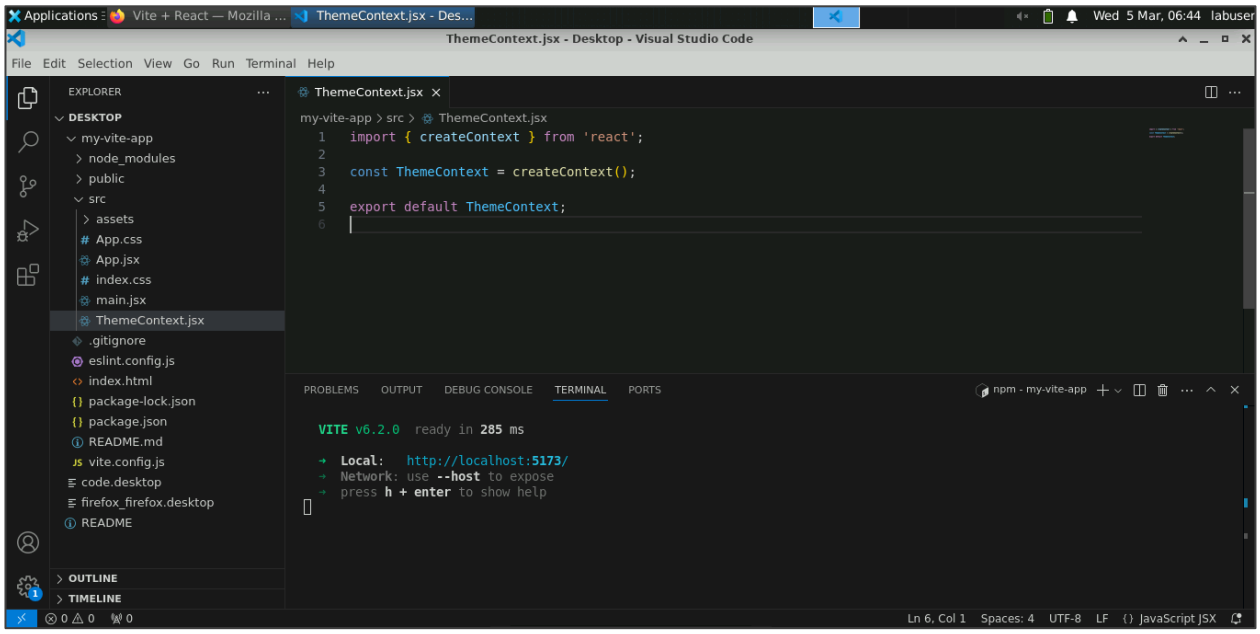
Step 2: Create react components

2.1 Inside the **src** folder, create a new file named **ThemeContext.jsx**

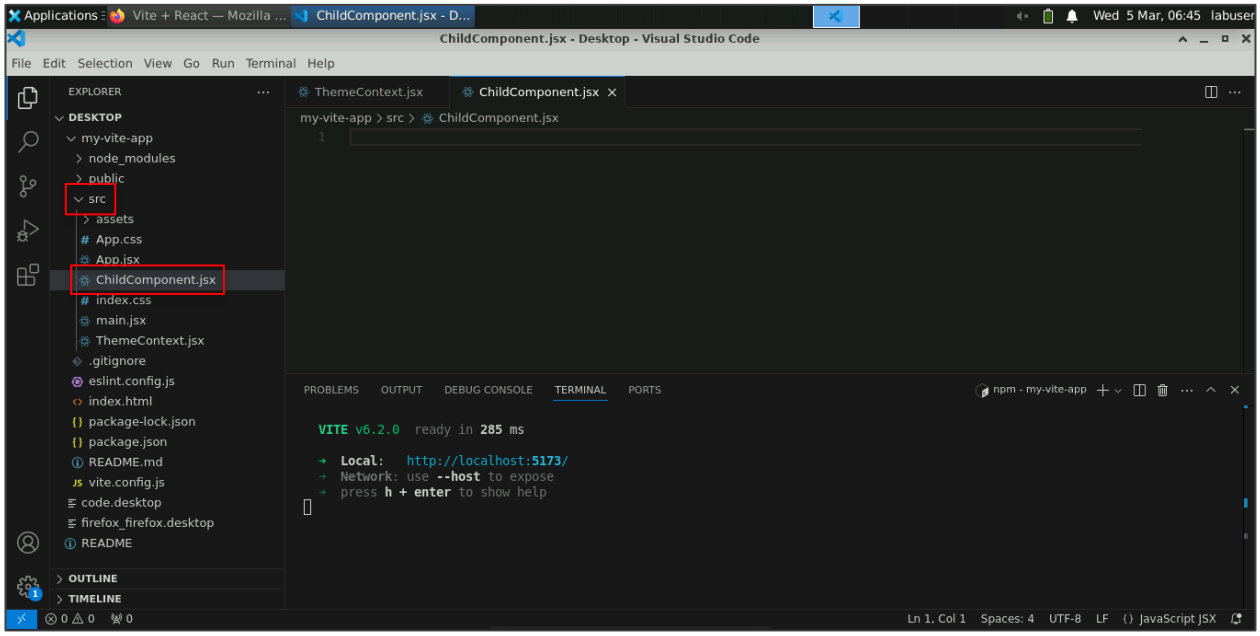


2.2 Add the following code to **ThemeContext.jsx** to create a context for theme management:

```
import { createContext } from 'react';

const ThemeContext = createContext();

export default ThemeContext;
```

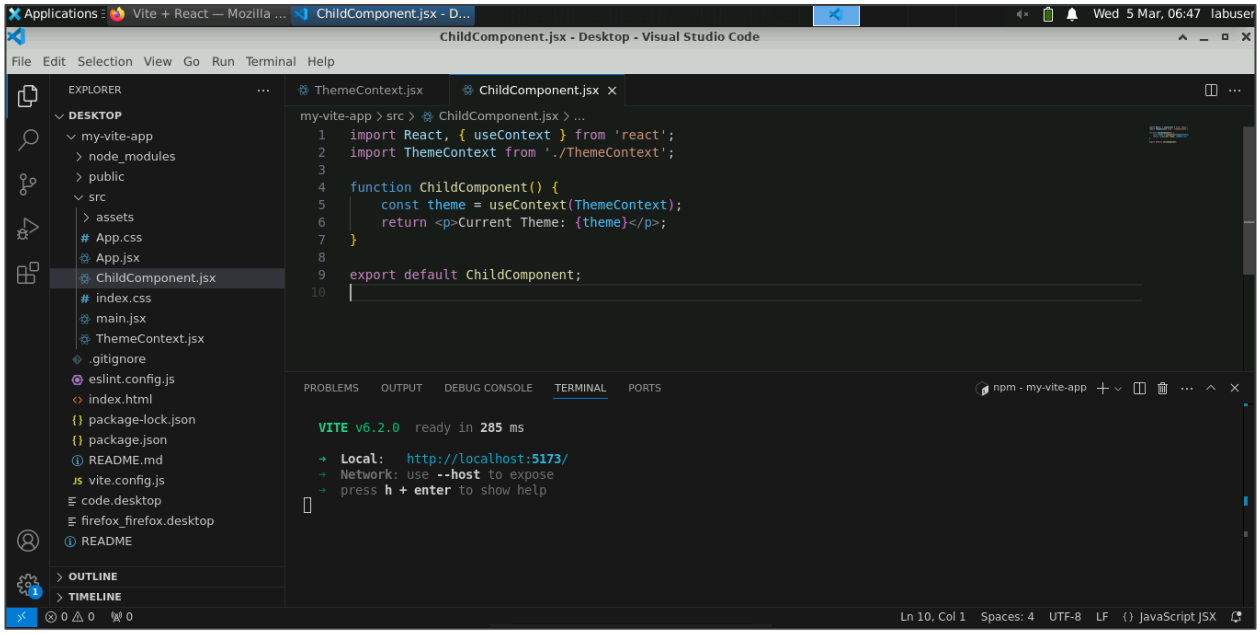


2.3 Inside the **src** folder, create a new file named **ChildComponent.jsx**



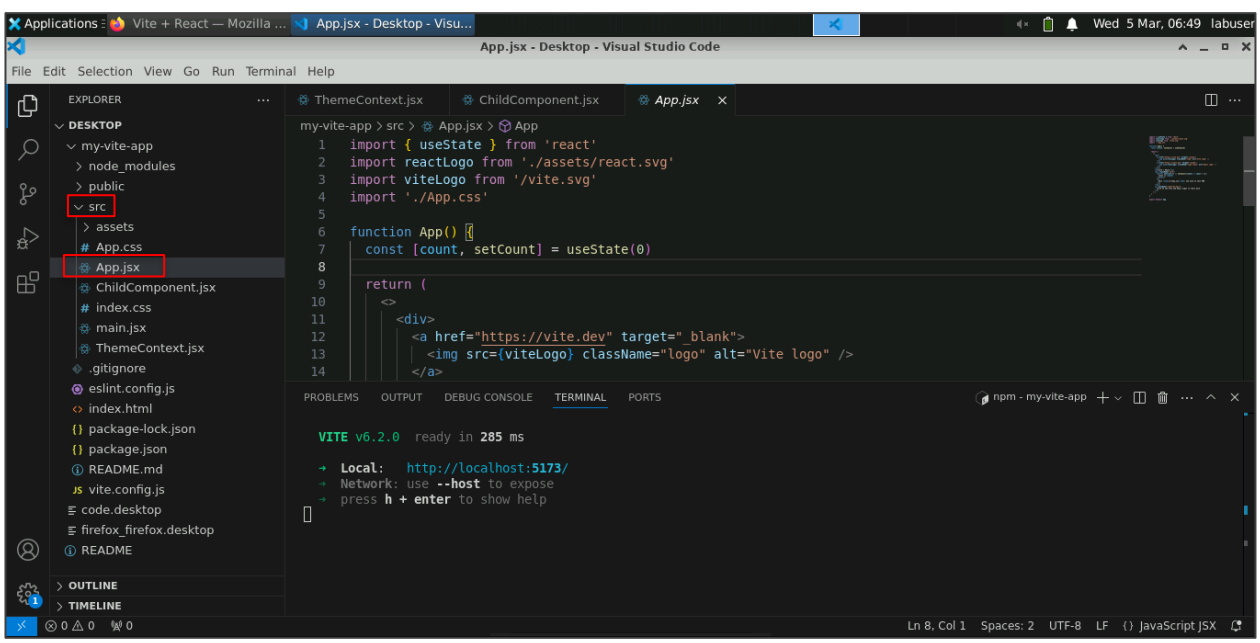
2.4 Add the following code to **ChildComponent.jsx** to consume the **ThemeContext** and display the current theme:

```
import React, { useContext } from 'react';
import ThemeContext from './ThemeContext';

function ChildComponent() {
   const theme = useContext(ThemeContext);
   return <p>Current Theme: {theme}</p>;
}

export default ChildComponent;
```



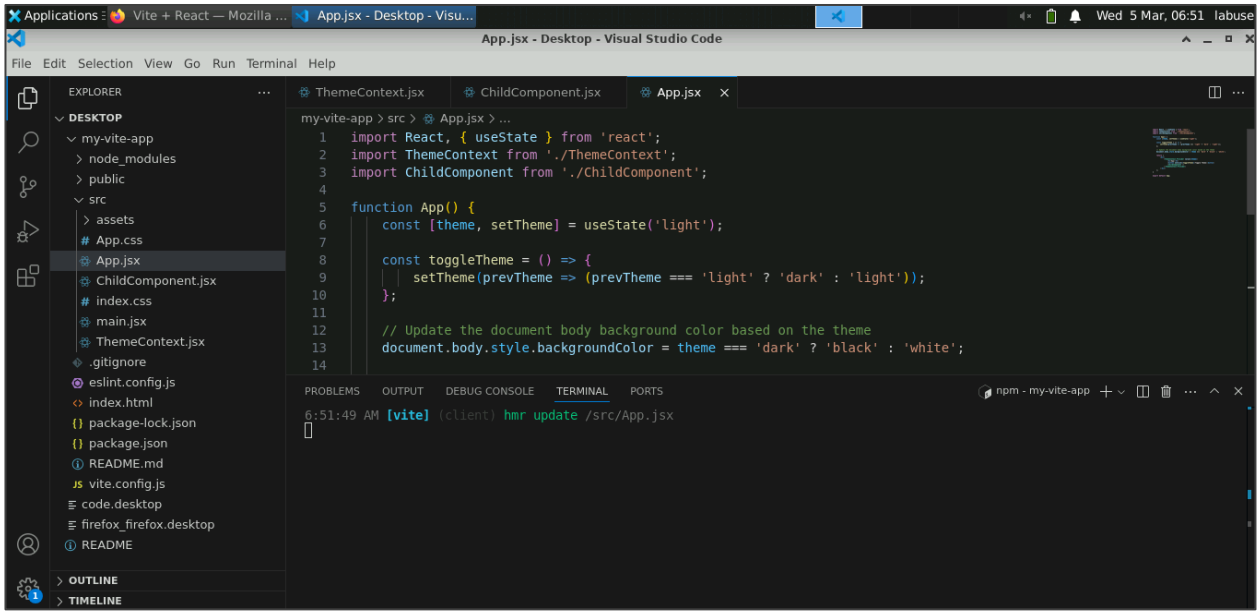
Step 3: Import and use the components

3.1 Open the **App.jsx** file inside the **src** folder

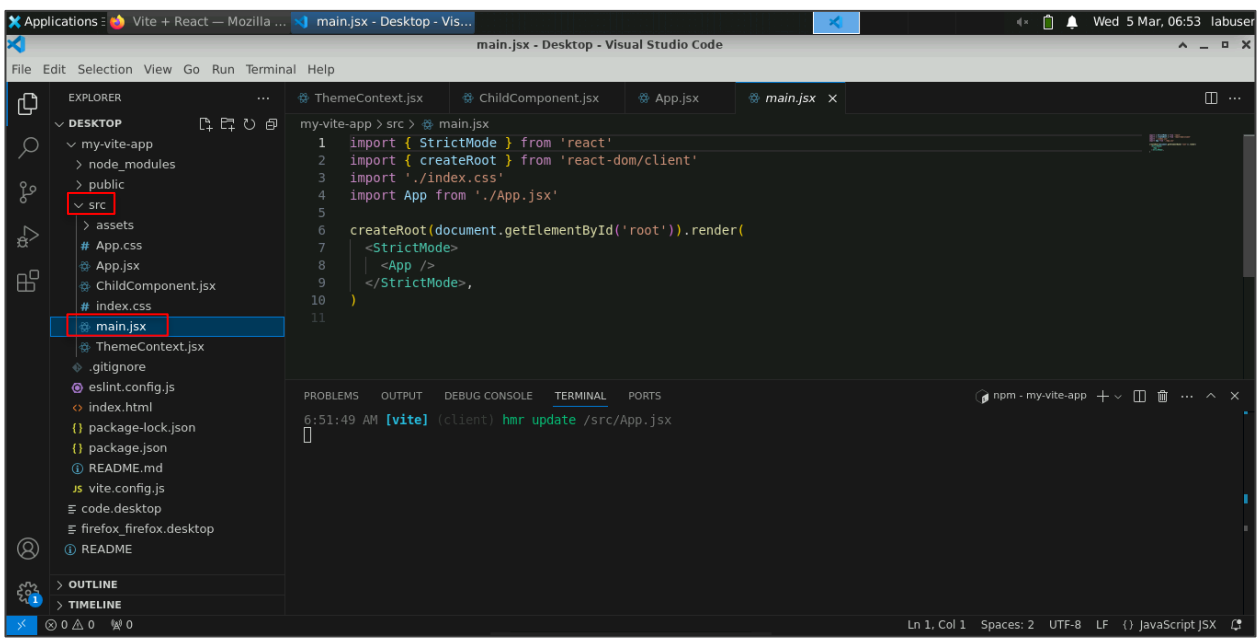


3.2 Paste the following code to include the new components in **App.jsx**:

```
import React, { useState } from 'react';
import ThemeContext from './ThemeContext';
import ChildComponent from './ChildComponent';

function App() {
  const [theme, setTheme] = useState('light');

  const toggleTheme = () => {
    setTheme(prevTheme => (prevTheme === 'light' ? 'dark' : 'light'));
  };

  // Update the document body background color based on the theme
  document.body.style.backgroundColor = theme === 'dark' ? 'black' : 'white';

  return (
    <div>
      <ThemeContext.Provider value={theme}>
        <h1>App</h1>
        <button onClick={toggleTheme}>Toggle Theme</button>
        <ChildComponent />
      </ThemeContext.Provider>
    </div>
  );
}

export default App;
```

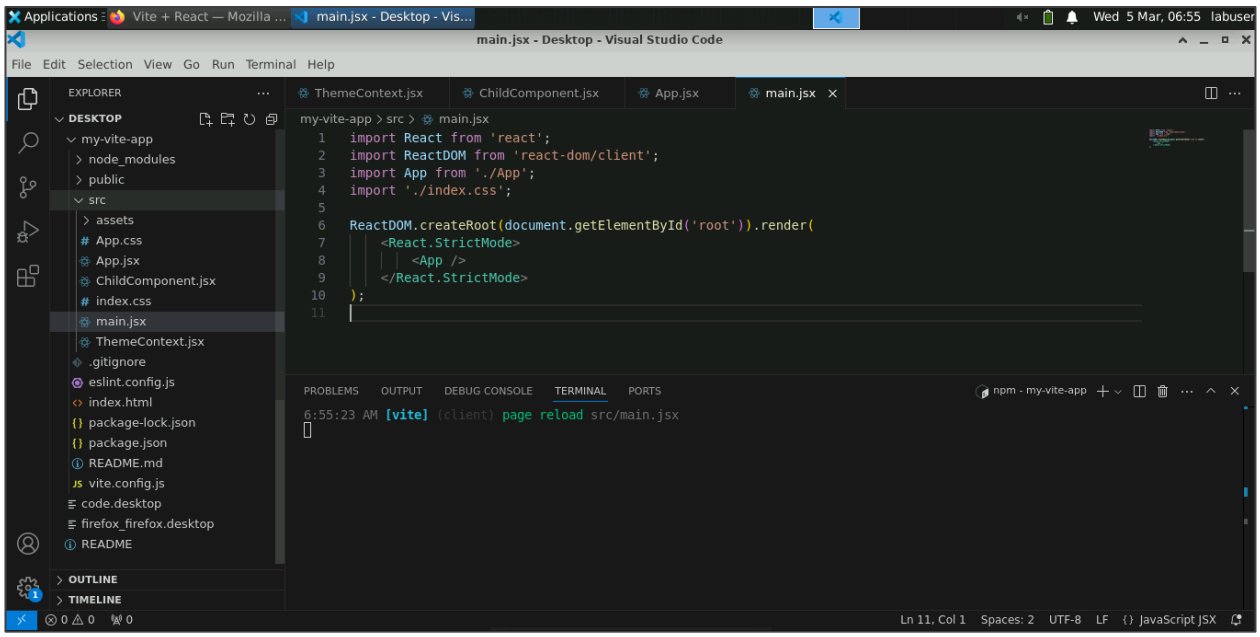


3.3 Navigate to the **src** folder and open the **main.jsx** file



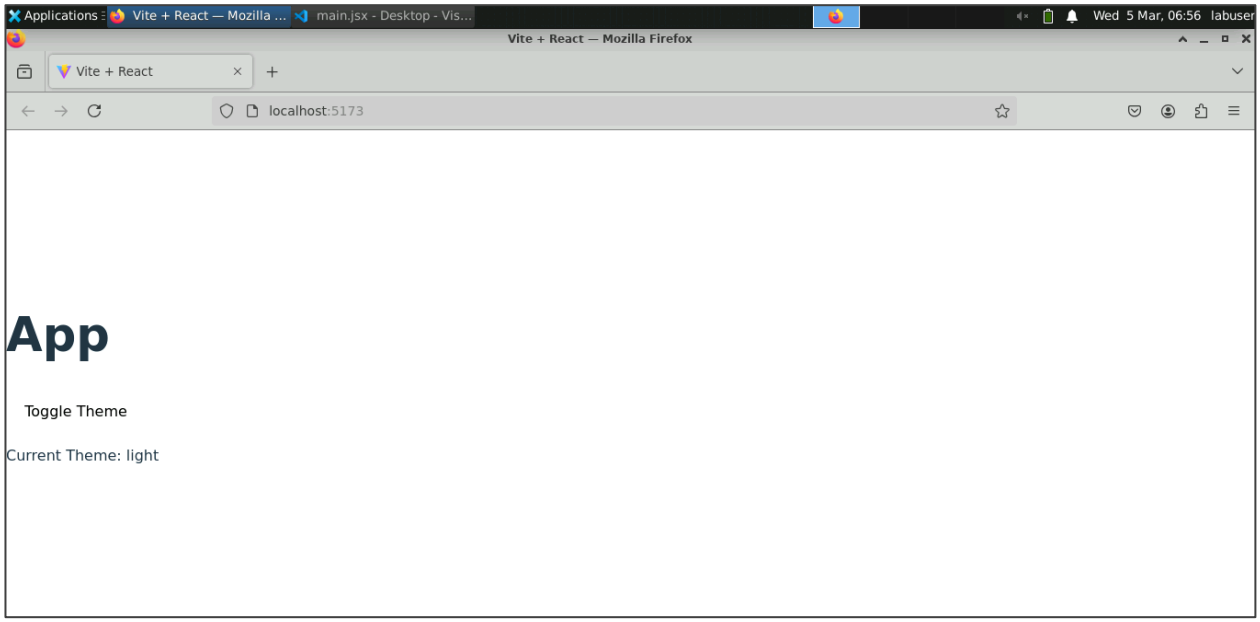
3.4 Paste the following code to include the new components in **main.jsx**:

```
import React from 'react';
import ReactDOM from 'react-dom/client';
import App from './App';
import './index.css';

ReactDOM.createRoot(document.getElementById('root')).render(
  <React.StrictMode>
    <App />
  </React.StrictMode>
);
```

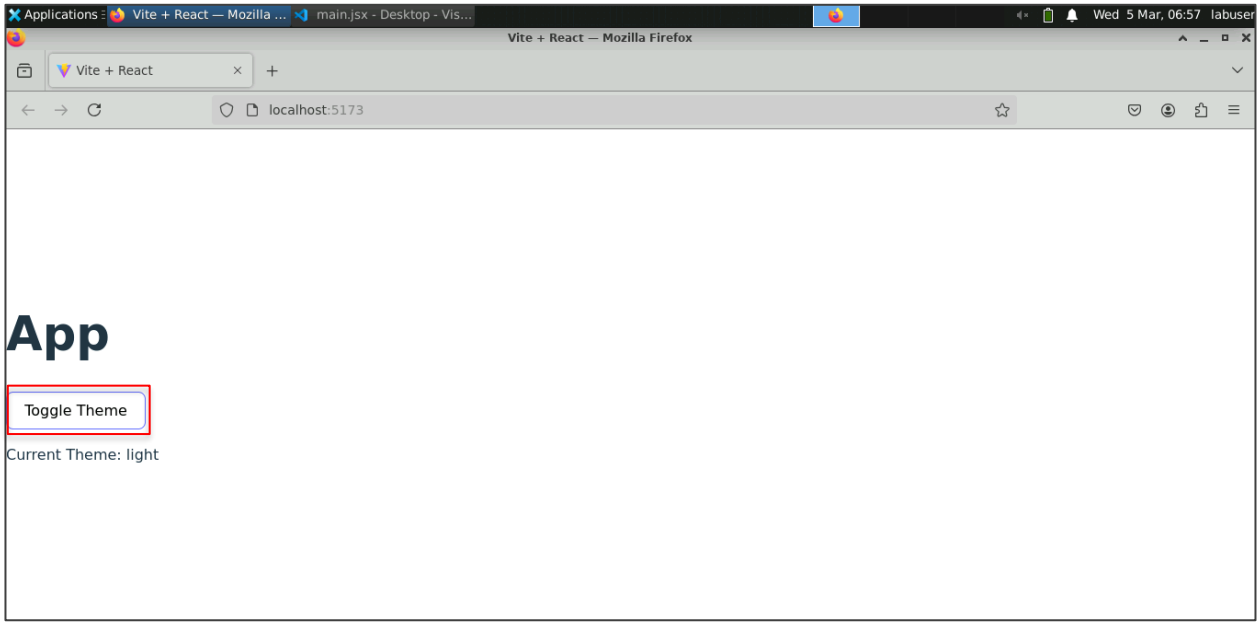


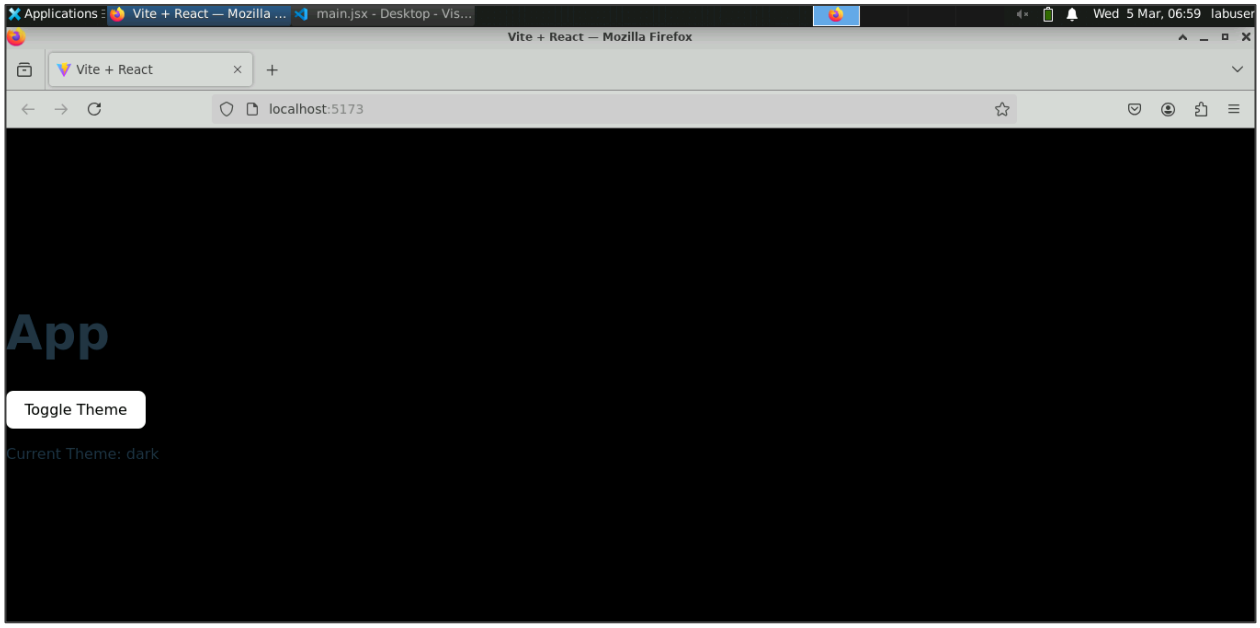
Step 4: Run and verify the application

4.1 Navigate to the browser and refresh the webpage



4.2 Click **Toggle Theme** to switch between light and dark themes





By following the above steps, you have successfully created a React application using Vite that utilizes the useContext Hook to manage a theme toggle feature. The application allows users to switch between light and dark themes dynamically, updating the background color accordingly.